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ORIGINAL PAPER

Uncertainty analysis of liquefaction-induced lateral spreading using fuzzy variables and genetic algorithm

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Abstract
One of the most destructive phenomena occurring during earthquakes is liquefaction-induced lateral spreading. So far, various prediction models have been proposed focusing only on the accuracy of the lateral spreading estimations. However, the reliability of predictive models is also essential since the uncertainties of input parameters result in uncertain estimates. To assess the performance of such models under uncertainty, the values of governing parameters from a well-known database are first fuzzified. Then, the extreme amounts of the liquified soil displacement are computed via coupling the genetic algorithm with the prediction models. This is accomplished through solving optimization problems of many objectives for analysis of the uncertainty effects by the fuzzy sets theory. Considering the fuzzy statistical indices, it is found that the slightly uncertain inputs can drastically influence the responses. Moreover, since the recent model tree (MT)-based predictive model is revealed to be vulnerable against uncertainty, a novel model named MT (New model) was developed. It is concluded that both the accuracy and reliability of the proposed relationships compare favorably with other available models.

Keywords Lateral spreading · Liquefaction · Uncertainty · Fuzzy theory

Introduction

Lateral spreading is one of the devastating phenomena affecting the layers of liquefiable soil during earthquakes. It refers to the movement of soil blocks on the liquefied layers due to seismic ground motion parallel to the earth surface. This phenomenon occurs in the areas of gentle slope (0 to 5%) and at free faces where the amount of movement varies from a few centimeters to several meters (Bartlett and Youd 1995). The lateral spreading caused significant damage in famous earthquakes such as San Francisco (1906), Niigata (1964), Alaska (1964), San Fernando (1971), Nihonkai-Chubu (1983), Superstition Hills (1987), Chi-Chi (1999), and Kocaeli (1999).

Predicting the amount of lateral spreading is essential to decrease its destructive effects (Jaimes et al. 2015). To this aim, many researchers have developed different models of various accuracy to predict lateral spreading. The proposed models usually do not take into account the uncertainty of the input parameters caused by the measurement errors, non-uniformity of soil characteristics, etc. Hence, in addition to the high accuracy, the reliability of the predictive equations is of utmost concern. Accordingly, the prediction model results should be evaluated on how they are influenced by the uncertainty of the input parameters.

In this study, first, the vulnerability of existing prediction models of lateral spreading (Dh) to the uncertainty of input parameters is investigated using the fuzzy sets theory and genetic algorithm (GA). Then, using the M5′ model tree algorithm (Witten and Wang 1997), a new model is presented for the estimation of the lateral spread in the free face and sloping ground conditions. This new set of equations is not only more accurate than the previous models but also is less vulnerable to the uncertainty of input parameters, so it is both accurate and reliable.

Lateral spreading models

Prediction approaches

To estimate the lateral spreading, researchers have employed several approaches: (a) simplified analytical methods such

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as the sliding blocks, the minimum potential energy, the viscous, the shear strength loss, and the strain re-hardening models (Bardet et al. 1999; Newmark 1965; Olson and Johnson 2008; Tokida et al. 1970; Towhata et al. 1992); (b) numerical methods including the finite element and finite difference approaches (e.g., Finn et al. 1994; Montgomery and Boulanger 2017; Yasuda et al. 1992); (c) experimental models like centrifuge and shaking table tests (Kutter et al. 2004; Sharp et al. 2003); and (d) empirical methods categorized to two types of (1) multiple linear regression (MLR) and (2) machine learning.

The current study focuses on the uncertainty analysis of data-driven models and the development of a new model based on a compiled dataset. Therefore, the previously developed empirical methods are reviewed below.

MLR

Some researchers have used the multiple linear regression method to predict the lateral spreading. The first empirical equation was proposed by Hamada et al. (1987) to predict lateral spreading using 60 data cases. The presented relationship was dependent only on the thickness of the liquefiable layer and the slope of the ground, and it was not so accurate to predict the lateral spreading at different sites. Youd and Perkins (1987) collected data from the California and Alaska earthquakes and estimated the lateral displacement using a liquefaction severity index (LSI), which depends on the moment magnitude of the earthquake and the shortest distance to the earthquake source. Bartlett and Youd (1995) used the multiple linear regression (MLR) analysis on 467 cases of past data to provide equations for predicting lateral spreading in both the sloping and free face conditions separately. Rauch and Martin III (2000) presented three equations with seismic, topography, and soil parameters, using the data of the sites considered by Bartlett and Youd (1992) and Bartlett and Youd (1995) plus at least 28 other sites. Using the data collected by Bartlett and Youd (1995), Bardet et al. (2002) proposed six relationships in which 6 input parameters were used. Youd et al. (2002) corrected the errors of the recorded lateral spreading data and also changed some of the input parameters. Accordingly, they introduced a new prediction model for the slope and free face conditions employing MLR.

Machine learning

This approach consists of different techniques for solving complex prediction problems (Derakhshani 2017; Jafariavval and Derakhshani 2020; Rahbarzare and Azadi 2019; Talebi and Derakhshani 2019). Using the artificial neural network (ANN), Wang and Rahman (1999) provided a model for predicting the lateral spreading with 367 data from 8 earthquakes. Baziar and Ghorbani (2005) employed the same method to develop a model to predict the ground lateral displacement for sloping and free face conditions. The accuracy of this model was higher than that of the MLR model of Youd et al. (2002). In 2006, Javadi et al. used genetic programming (GP) to create separate equations for the slope and free face conditions, which was more accurate than previous models. Baziar and Saeedi Azizkandi (2013) combined ANN and GP to estimate lateral spreading. They provided a predictive relationship for both sloping and free face ground conditions and evaluated its behavior via parametric study. Goh and Zhang (2014) offered a new model using multivariate adaptive regression splines (MARS) approach, which was highly accurate but very complex. Avval and Derakhshani (2019) proposed a set of formulae for predicting lateral spreading in both free face and slope conditions, through the M5′ model tree technique. This equation was superior in accuracy and simplicity compared to all previous models. Some of the relationships presented for predicting the lateral spreading are exhibited in Table 1.

This research implies the development of a new predictive model via M5′ model tree approach which compares favorably with previous models in terms of both accuracy and reliability.

Reliability analysis approaches

The main purpose of the abovementioned studies was to achieve high estimation accuracy, while the effect of input data uncertainties and their influence on the reliability of model results, have been rarely considered.

Al Bawwab (2005) collected and filtered the data of various case histories of lateral spreading and created a set of probabilistic models via maximum likelihood, linear, and nonlinear regression assessment methods. In addition, different sources of uncertainties were considered in the prediction of the amount of the lateral ground deformations.

Kalantary et al. (2013) suggested a robust optimization approach for assessing the influence of uncertainty of each parameter on the output of a new lateral spreading model. They showed that their model outperforms Al Bawwab model for datasets involving uncertainty.

Franke and Kramer (2014) proposed a procedure to yield performance-based estimates of lateral spread displacement considering the well-known empirical model of Youd et al. (2002). The predictive equations were modified based on the principles of probabilistic seismic hazard analysis.

Jafarian and Nasri (2016) refined the famous database of lateral ground displacement compiled by Youd et al. (2002) and proposed a new empirical model using a regression technique. They also employed various statistical and probabilistic approaches such as the first-order reliability method (FORM), the second-order reliability method (SORM), and

Table. 1 Examples of formulae presented to estimate the lateral spreading

Author	Method	Condition	Lateral spreading formulas
Bartlett and Youd (1995)	**MLR**	Sloping ground	$\log(Dh) = -15.787 + 1.178\,M - 0.927\,\log R - 0.013\,R + 0.429\,\log S + 0.348\,\log T_{15} + 4.527\,\log\left(100 - F_{15}\right) - 0.922\,D50_{15}$
		Free face	$\log(Dh) = -16.366 + 1.178\,M - 0.927\log R - 0.013\,R + 0.657\log W + 0.348\log T_{15} + 4.527\log\left(100 - F_{15}\right) - 0.922 D50_{15}$
Youd et al. (2002)	**MLR**	Sloping ground	$R^* = R_0 + R, R_0 = 10^{(0.89\,M-5.64)}$ $\log(Dh) = -16.213 + 1.532\,M - 1.406\log R^* - 0.012\,R + 0.338\log S + 0.54\log T_{15} + 3.413\log\left(100 - F_{15}\right) - 0.795\log\left(D50_{15} + 0.1\right)$
		Free face	$\log Dh = -16.713 + 1.532M - 1.406\log\left(R^*\right) - 0.012R + 0.592\log W + 0.54\log T_{15} + 3.413\log\left(100 - F_{15}\right) - 0.795\log\left(D50_{15} + 0.1mm\right)$
Javadi et al. (2006)	**GP**	Sloping ground	$Dh_c = 0.8\frac{F_{15}}{M} + 0.0014{F_{15}}^2 + 0.16T_{15} + 0.112S + 0.04\frac{ST_{15}}{D50_{15}} - 0.026RD50_{15} + 1.14$
		Free face	$Dh_c = -163.1\frac{1}{M^2} + 57\frac{1}{RF_{15}} - 0.0035\frac{{T_{15}}^2}{W{D50_{15}}^2} + 0.02\frac{{T_{15}}^2}{F_{15}{D50_{15}}^2} - 0.26\frac{{T_{15}}^2}{{F_{15}}^2} + 0.006{T_{15}}^2 - 0.0013W^2 + 0.0002M^2WT_{15} + 3.7$
Baziar and Saeedi Azizkandi (2013)	**ANN and GP**	Sloping ground and free face	$Dh_c = \frac{54.36T_{15}}{D50_{15} + 0.6532} - \frac{55.34T_{15}}{D50_{15} + 0.6689} + \frac{196.9T_{15}}{W + 0.9212} - \frac{199.8T_{15}}{W + 0.9434} + \frac{0.0446(W - S)}{R} - \frac{1.718}{S + 0.8956} - 0.02452T_{15}F_{15} - 0.00625F_{15}S + 0.001474R\left(W - T_{15}\right) - 0.06875T_{15}(W - S) + M\left(0.1058T_{15} + 0.009652T_{15}W - 0.1225\right) + 0.00024T_{15}{F_{15}}^2 - 0.00255RWS + 2.6$
Avval and Derakhshani (2019)	**MT**	Sloping ground and free face	If $F_{15} \geq 20.02$ $Dh_c = \frac{M^{18.025}W^{0.255}S^{0.03}{T_{15}}^{0.353}\left(100-F_{15}\right)^{3.821}}{R^{1.274}10^{22.38}\left(D50_{15}+0.1\right)^{0.609}}$
			If $F_{15} < 20.02$ and $W \leq 4.529$ and $T_{15} \leq 9.14$ $Dh_c = \frac{M^{1.754}S^{0.282}{T_{15}}^{0.308}\left(100-F_{15}\right)^{1.909}}{R^{0.123}10^{5.215}\left(D50_{15}+0.1\right)^{0.32}W^{0.197}}$
			If $F_{15} < 20.02$ and $W \leq 4.529$ and $T_{15} > 9.14$ $Dh_c = \frac{M^{1.754}S^{0.275}{T_{15}}^{0.105}\left(100-F_{15}\right)^{7.859}}{R^{0.123}10^{17.024}\left(D50_{15}+0.1\right)^{1.613}W^{0.389}}$
			If $F_{15} < 20.02$ and $W > 4.529$ and $D50_{15} \leq 0.437$ $Dh_c = \frac{M^{5.668}W^{0.425}S^{0.038}{T_{15}}^{0.535}\left(100-F_{15}\right)^{6.187}}{R^{0.407}10^{17.373}\left(D50_{15}+0.1\right)^{1.135}}$
			If $F_{15} < 20.02$ and $W > 4.529$ and $D50_{15} > 0.437$ $Dh_c = \frac{M^{8.695}W^{0.228}S^{0.038}{T_{15}}^{0.353}\left(100-F_{15}\right)^{2.893}}{R^{0.628}10^{12.943}\left(D50_{15}+0.1\right)^{0.507}}$

the Monte Carlo simulation (MCS) method to account for the uncertainty of governing parameters.

A different less-employed approach to assess the reliability in civil engineering problems is the application of the fuzzy method (Razifard et al. 2019). Some examples of the applications of this approach are as follows. Branisavljevic and Ivetic (2006), Haghighi and Keramat (2012), and Haghighi and Asl (2014) evaluated the effects of uncertainties on the pipe pressure and discharge in water distribution networks in different areas using the fuzzy logic. Gong et al. (2015), Haghighi and Ayati (2016), and Derakhshani (2018) conducted uncertainty analysis on the design of ground slope, gravity dam and suction caisson using the fuzzy method. They showed that the uncertainty of input parameters may result in significant changes in system response and the design is unreliable without consideration of uncertainties.

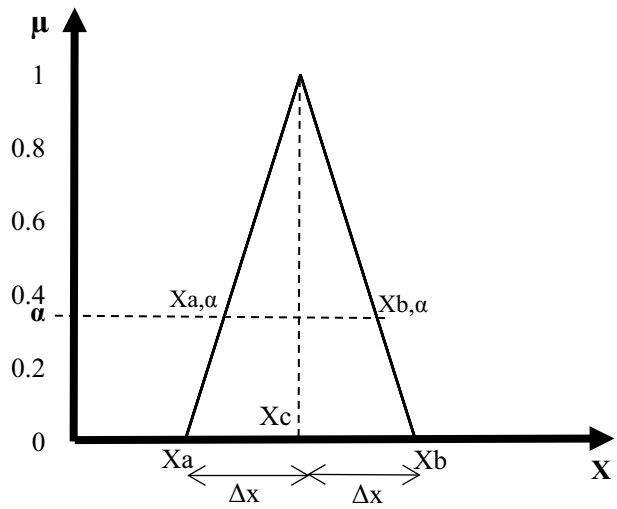


Fig. 1 Triangular membership function

This paper concentrates on a robust approach based on the fuzzy sets theory, to analyze the influence of the uncertainties of the governing parameters on the lateral spreading estimation. Generally, the accurate values of these parameters cannot be determined due to the limited test data available in many geotechnical projects. This results in uncertain calculated lateral spreading, which makes it difficult to evaluate if the estimated responses are excessive as compared to the allowable ground deformation. Where the data are lacking or inadequate and it is not possible to fully define a probability distribution, a triangular fuzzy number can be utilized to model uncertain soil parameters for uncertainty analysis. A triangular fuzzy membership function can be defined corresponding to the estimation of the highest, the lowest and the most probable values (upper bound, lower bound, and mode) of a soil parameter. To analyze the effects of uncertainty on the lateral spreading, all the uncertain soil parameters are introduced by fuzzy numbers and different levels of uncertainty are considered by the alpha-cut method.

Uncertainty analysis

Application of fuzzy theory

In engineering systems, uncertainty is divided into two types of aleatory and epistemic. The aleatory uncertainty is due to probabilistic variability and cannot be reduced, but the Epistemic uncertainty can be reduced due to its relevance to human knowledge. The input parameters used for the prediction of the lateral spreading are subject to measurement error and uncertainty. A reliable predictive model necessarily accounts for these uncertainties. In this study, triangular membership functions based on the fuzzy set theory are used to evaluate the uncertainty influence on the results of the prediction models.

In the fuzzy set theory, the uncertainties are represented by fuzzy numbers where the function μ (X) determines the degree of membership of parameter x in the corresponding set and its value is between 0 and 1. Figure 1 illustrates a triangular membership function in which X_c refers to the measured value of the parameter, X_a is the lower bound and X_b is the upper bound of X_c due to the maximum uncertainty of ΔX. As shown in Fig. 1, an α-cut is applied to the fuzzy number where an interval is defined as $[X_{a,\alpha}, X_{b,\alpha}]$. When $\alpha = 0$, the corresponding interval i.e. $[X_a, X_b]$ is referred to as "support." Crisp value (X_c) or the most likely value of the fuzzy number corresponds to $\alpha = 1$.

For each input parameter, the values in the well-known compiled database of Youd et al. (2002) employed in creating different models for the prediction of lateral spreading, are considered as the X_c. Then, every parameter is represented by a triangular fuzzy number with the maximum uncertainty of ΔX, followed by the discretization using some α-cuts. Therefore, some intervals are defined for each parameter that reflect the levels of the input uncertainties. To understand the influence of input fuzziness on the model results, the extreme values of lateral spreading must be found. For this purpose, the nonlinear predictive

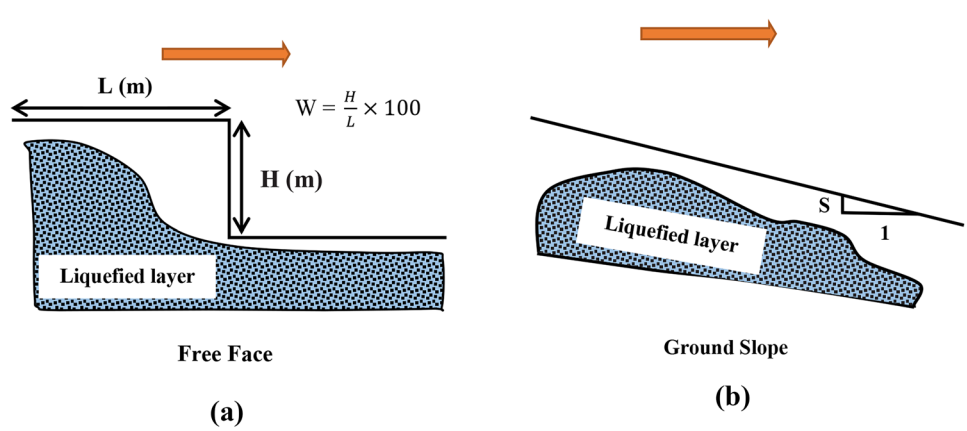


Fig. 2 The topography of lateral spreading sites. (**a**) Free face. (**b**) Ground slope

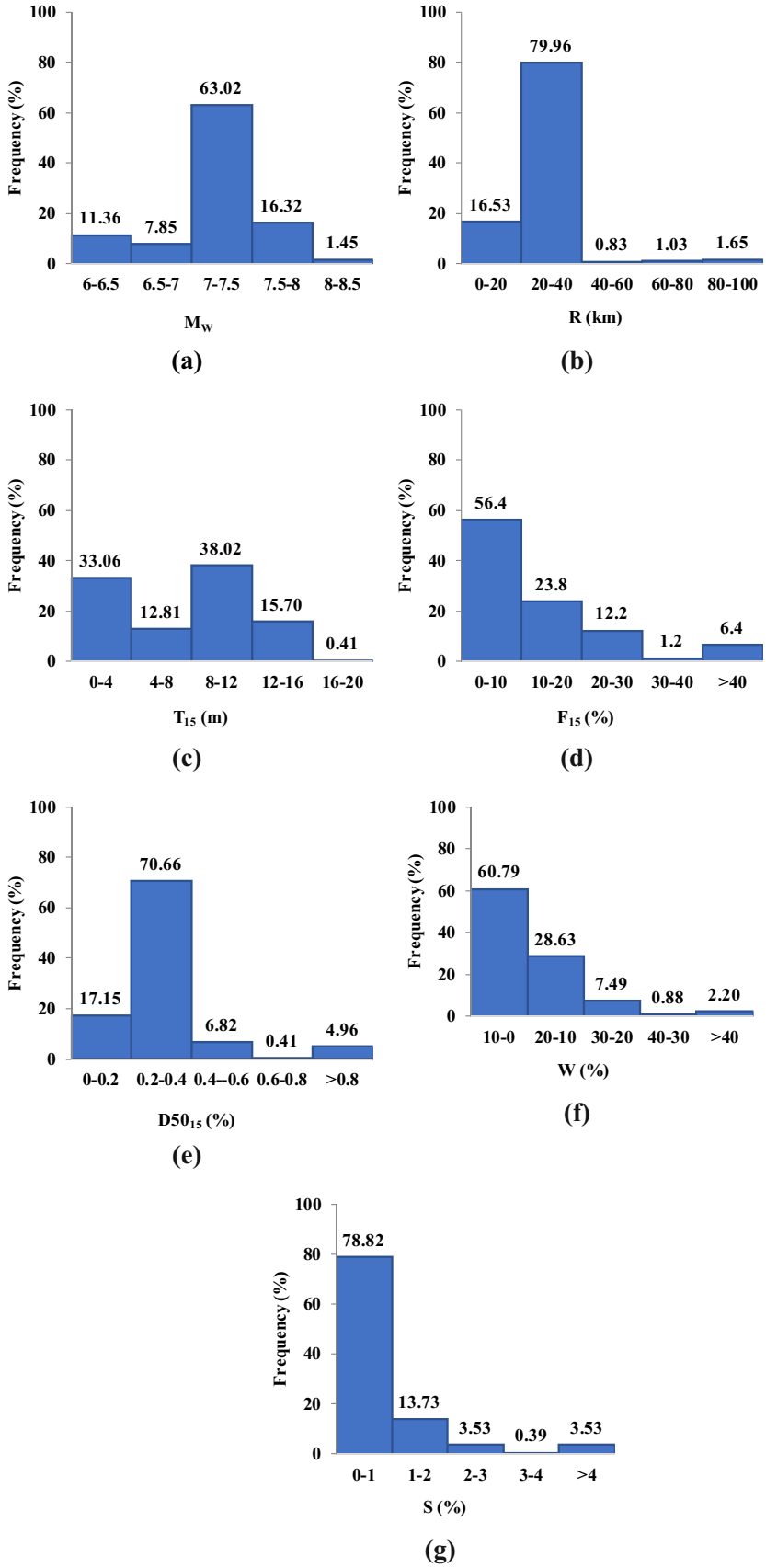


Fig. 3 Histograms of input variables. (**a**) Moment magnitude. (**b**) Least distance to the seismic energy source. (**c**) Cumulative thickness of saturated granular layers with corrected blow counts of SPT lower than 15. (**d**) Average fines content for granular materials within T_{15}. (**e**) Average mean size of granular materials within T_{15}. (**f**) Ratio of the free face. (**g**) Slope of the ground

models should be once minimized and once maximized at each level of uncertainty. Hence, an optimization problem is formed with the objective function of the lateral spreading estimation models to be minimized and also maximized. In other words, for fuzzy analysis of different lateral spreading prediction models, a single-objective optimization is accomplished twice at each α-cut. In this paper, first, the input parameters are fuzzified through a subroutine coupled to the functions estimating the amount of lateral spreading. Numerous optimization problems of various objectives are then created for the defined α-cuts. Accordingly, the estimation model is linked with the genetic algorithm (GA) to yield the corresponding fuzzy intervals (extreme values) of the lateral spreading.

Genetic algorithm

The genetic algorithm is one of the most famous metaheuristic optimization algorithms inspired by the Darwin's theory of evolution (Holland 1992). In this approach, a group of potential solutions changes step by step to obtain the optimum solution. The population size is the number of random solutions and the generations are the iterations of optimization. The genetic algorithm starts with a number of chromosomes (primary population) made up of discrete units of the gene (Kuroki et al. 2010). Among the existing chromosomes, more suitable ones are selected as parents and then crossover and mutation operators are used to reproduce a new chromosome (child). The repeated application of the mentioned operations gradually leads to the creation of new better generations. The space of chromosomes is searched for the one that is capable of optimizing the fitness function (criterion for the goodness of a solution). The process is terminated based on a specific number of generations produced or a suitable fitness obtained.

Database

This study uses data of 484 cases of lateral spreading that occurred during the seismic events of Japan, USA, Turkey, and Taiwan reported by Youd et al. (2002). The effective parameters on the amount of lateral spreading displacement (Dh) recorded in this dataset are Mw, moment magnitude; R (km), distance to the earthquake source; T_{15} (m), thickness of layers with the standard penetration test (SPT) number lower than 15 ($N_{SPT} \leq 15$); F_{15} (%), average fine content in T_{15}; $D50_{15}$ (mm), average grain size in T15 and the parameters of topography as demonstrated in Fig. 2 including S (%), slope of the ground, and W (%), ratio of the free face.

The histogram plots of the governing parameters are depicted in Fig. 3. In addition, statistical description (minimum, maximum, average, and standard deviation) of these variables based on the values reported in the database is listed in Table 2. This dataset is employed to assess the vulnerability of the predictive equations to the uncertainty as well as the development of a new prediction model.

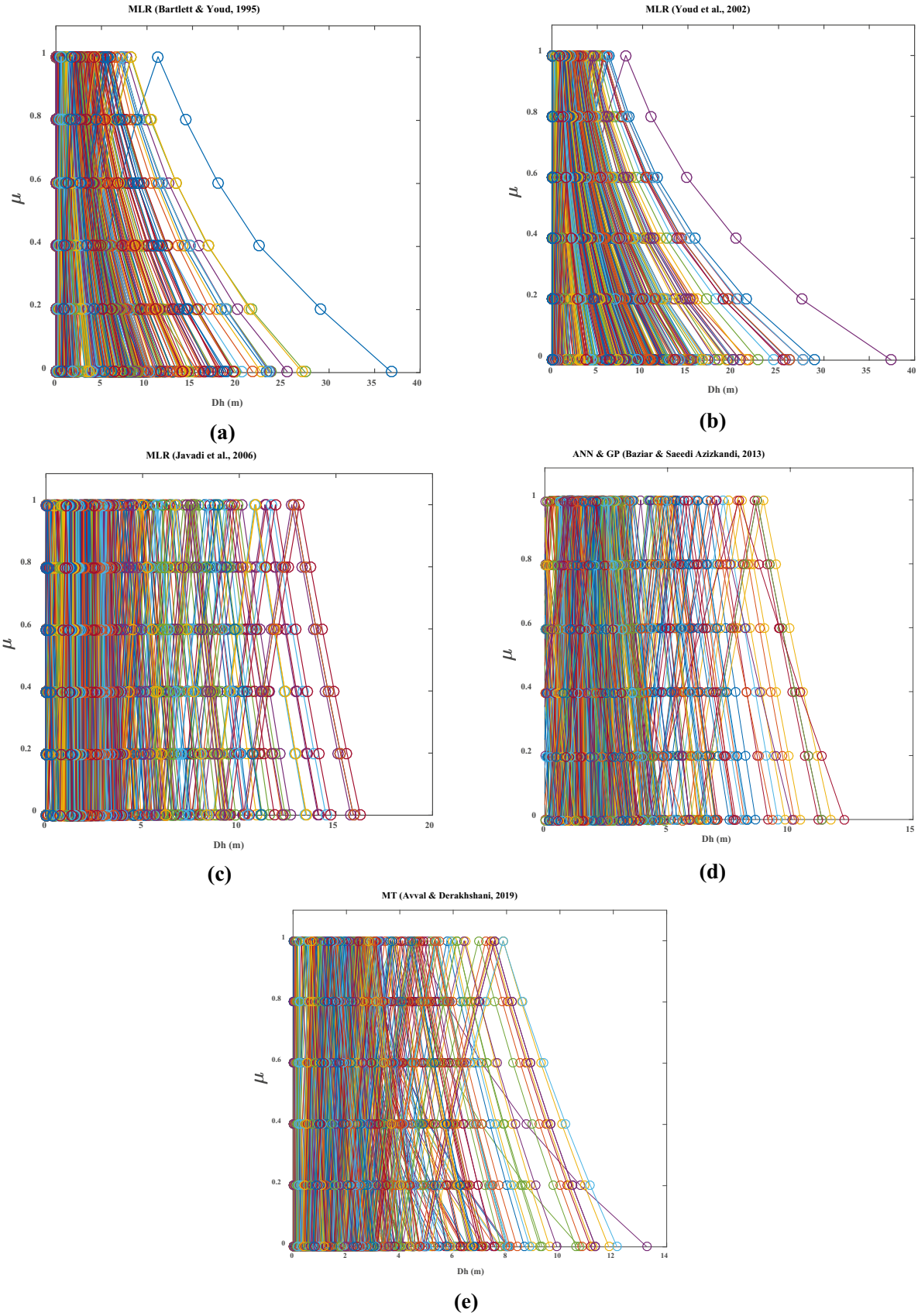
Fig. 4 Fuzzy responses of various models for all data. (**a**) MLR (Bartlett and Youd 1995). (**b**) MLR (Youd et al. 2002). (**c**) MLR (Javadi et al. 2006). (**d**) ANN and GP (Baziar and Saeedi Azizkandi 2013). (**e**) MT (Avval and Derakhshani 2019) ▶

Determining the uncertainty of different models

The reliability of previously presented predictive relationships (Table 1) is evaluated as follows. First, using the available data, the values of lateral displacement are calculated via the abovementioned relationships (the crisp values). Then, using the triangular membership functions, the uncertainties of ±5%, ±4%, ±3%, ±2%, and ±1% are applied to the input parameters. The mentioned levels of uncertainty correspond to the α-cuts of 0, 0.2, 0.4, 0.6, and 0.8. In the membership function triangle, α = 0 refers to the maximum uncertainty at the support and α = 1 indicates the crisp (most probable) value of the function. Considering the continuous intervals of uncertainty for all input parameters, different combinations of the inputs with different uncertainties can be selected by an optimization method like the genetic algorithm (GA) to find the maximum and minimum amounts of the lateral displacement. In other words, for each α-cut, there are two 7-dimensional optimization problems of minimization and maximization of lateral spreading to be solved by the GA. The fuzzy sets of the responses of the previous models such as MLR (Bartlett and Youd 1995; Javadi et al. 2006; Youd et al. 2002), ANN and GP (Baziar and Saeedi Azizkandi 2013), and MT (Avval and Derakhshani 2019) for all data are illustrated in Fig. 4. To determine the overall effect of uncertainty on different models, statistical indicators such as correlation coefficient (CC), root mean square error (RMSE), and mean absolute error (MAE) are calculated. The formulae of these statistical indicators are presented by Eqs. 1 to 3.

Table. 2 Statistics of all data

	M	R (km)	T_{15}(m)	F_{15}(%)	$D50_{15}$(mm)	W (%)	S (%)	Dh (m)
Min	6.40	0.20	0.01	0.00	0.04	0.00	0.00	0.01
Max	9.20	100.00	19.70	70.00	12.00	56.80	11.00	10.159
Average	7.38	21.25	7.51	13.01	0.40	5.02	0.50	2.15
Standard deviation	0.452	13.760	4.357	12.576	0.762	8.139	1.202	1.758


MLR (Bartlett & Youd, 1995)
MLR (Youd et al., 2002)
MLR (Javadi et al., 2006)
ANN & GP (Baziar & Saeedi Azizkandi, 2013)
MT (Avval & Derakhshani, 2019)
μ
Dh (m)
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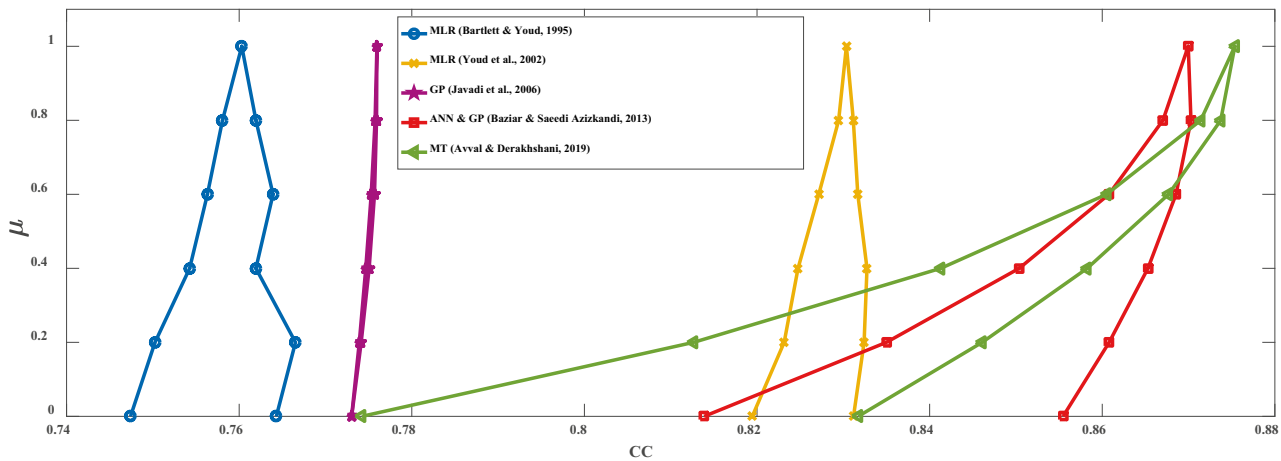

Fig. 5 Fuzzy sets of the correlation coefficients of different models' outputs

$$\mathrm{CC} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2 \sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{\left(x_i - y_i\right)^2}{n}} \tag{2}$$

$$\mathrm{MAE} = \frac{\sum_{\mathrm{i}=1}^{\mathrm{n}}\left|\mathrm{x_i} - \mathrm{y_i}\right|}{\mathrm{n}} \tag{3}$$

where x_i and y_i are the values of measured and predicted lateral displacement, respectively, $\bar{x}$ and $\bar{y}$ indicate the average values of measurement and prediction data, and n represents the total number of data.

Figure 5 exhibits the membership functions of the correlation coefficients calculated based on the results of the models presented in Table 1. As can be seen, the correlation coefficients of all predictive models vary between 0.74 and 0.88. It is noteworthy that the crisp values of the triangular membership functions represent the deterministic results (without considering the uncertainty), while the supports show the extent to which the models are influenced by the uncertainty. In other words, the greater support of the fuzzy triangle indicates that the model is more vulnerable to the effects of the input uncertainties. As shown in Fig. 5, the support lengths of the predictive models of Bartlett and Youd (1995), Baziar and Saeedi Azizkandi (2013), Javadi et al. (2006), Youd et al. (2002), and Avval and Derakhshani (2019) are about 0.017, 0.012, 0, 0.042, and 0.058, respectively. As can be seen, although the model of Avval and Derakhshani (2019) with the crisp CC = 0.875 has the highest accuracy among all models, its support is larger than other models indicating the greatest vulnerability to the effects of uncertainty.

Figures 6 and 7 illustrate the membership functions of the root mean squared error and the mean absolute error of the relationships given in Table 1, respectively. The support sizes of the membership functions of the models of Bartlett and Youd (1995) and Youd et al. (2002) are about 4 and 5.1 for RMSE, and 3.3 and 4.4 for MAE. In both figures, these models are more influenced by the uncertainty compared to

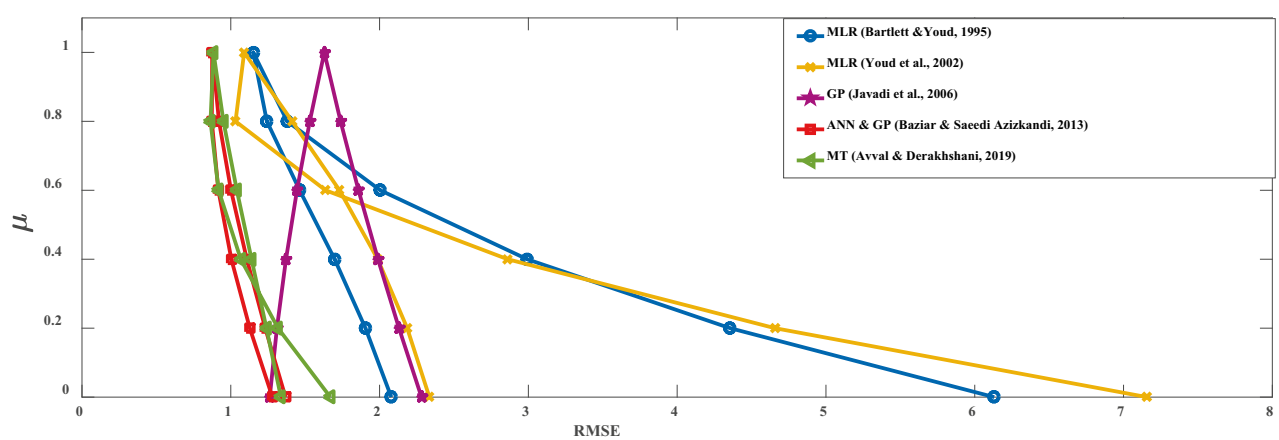

Fig. 6 Fuzzy sets of the root mean squared errors of different models' outputs

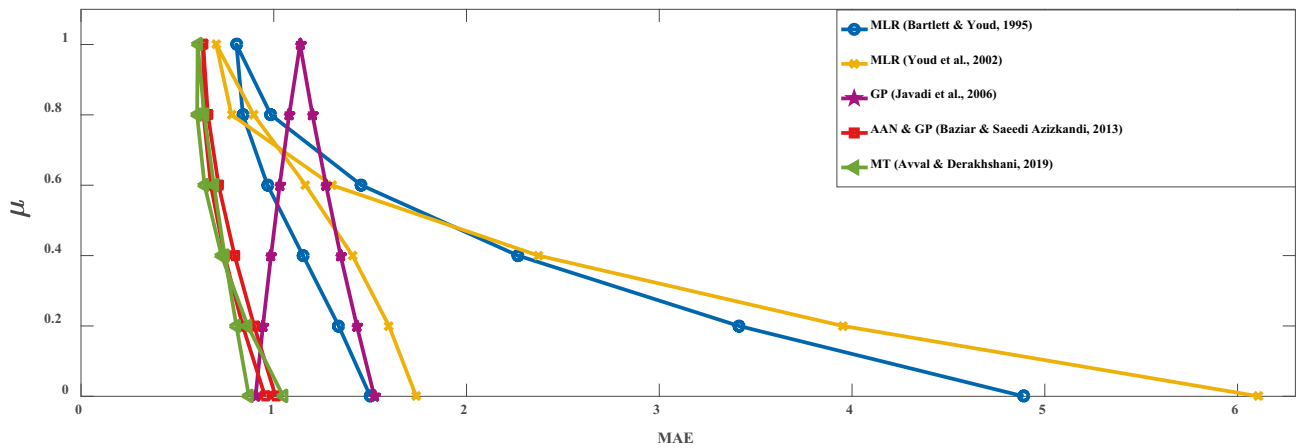

Fig. 7 Fuzzy sets of the mean absolute errors of different models' outputs

the other models. Also, according to Figs. 6 and 7, the models of Avval and Derakhshani (2019) and Baziar and Saeedi Azizkandi (2013) are less vulnerable to input uncertainties considering the RMSE and MAE fuzzy sets.

Development of an enhanced model for predicting the lateral spreading

As mentioned above, the previous predictive models aimed at achieving high accuracy while they are vulnerable to the effects of input uncertainties. Specifically, the MT model (Avval and Derakhshani 2019) is more accurate than the other models considering the crisp values of statistical indices; however, the fuzzy set of its correlation coefficient indicates that it is not so reliable. Hence, an attempt is made to create a novel model, called "MT (New model)," that is as accurate as the previous MT model, but less sensitive to the uncertainty.

MT

MT is a data mining method used for classification and prediction purposes (Khorrami and Derakhshani 2019; Khorrami et al. 2020; Khoshroo and Derakhshani 2020). The model tree algorithm first introduced by Quinlan (1992) is called M5, and later, it was improved to M5′ by Witten and Wang (1997). They reimplemented the M5 method with modifications to create models which are compact, performant, and capable of dealing with missing values. This algorithm is capable of dividing complex problems into smaller parts of easier problems. It mainly comprises making a tree and extracting knowledge from it.

M5' modeling is carried out through processes of making, pruning, and smoothing and yields a tree-like structure with root, branches, and leaves (Fig. 8). At first, the data is divided into several parts, then, a multivariate linear regression model is fitted to each part of the data. The criterion for dividing the data (corresponding to the branches) is to maximize the reduction of the standard deviation of the subsets.

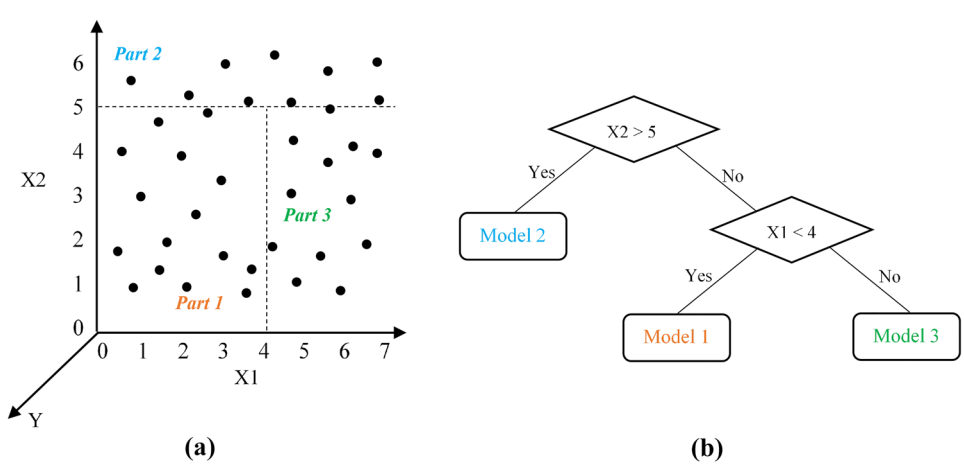

Fig. 8 Example of estimation using the M5′ model tree. (**a**) Subsets of input parameter space (X1*X2). (**b**) Created tree and the models for the subsets

Table. 3 Statistics of the train and test data used to build the new MT model

Parameters	Minimum		Maximum		Mean		Standard deviation	
	Training	Testing	Training	Testing	Training	Testing	Training	Testing
M	6.4	6.4	9.2	9.2	7.39	7.35	0.45	0.44
R (km)	0.2	0.2	100	80	21.48	19.63	13.75	11.23
T_{15} (m)	0.01	0.2	19.7	16.7	7.46	7.69	4.35	4.34
F_{15} (%)	0	0	68	70	12.99	13.54	12.68	12.94
$D50_{15}$ (mm)	0.06	0.04	12.00	12.0	0.4	0.47	0.82	1.22
W (%)	1	1	55.68	56.8	5.56	5.43	7.81	7.89
S (%)	0.05	0.1	11	11	0.96	1.04	0.98	1.5

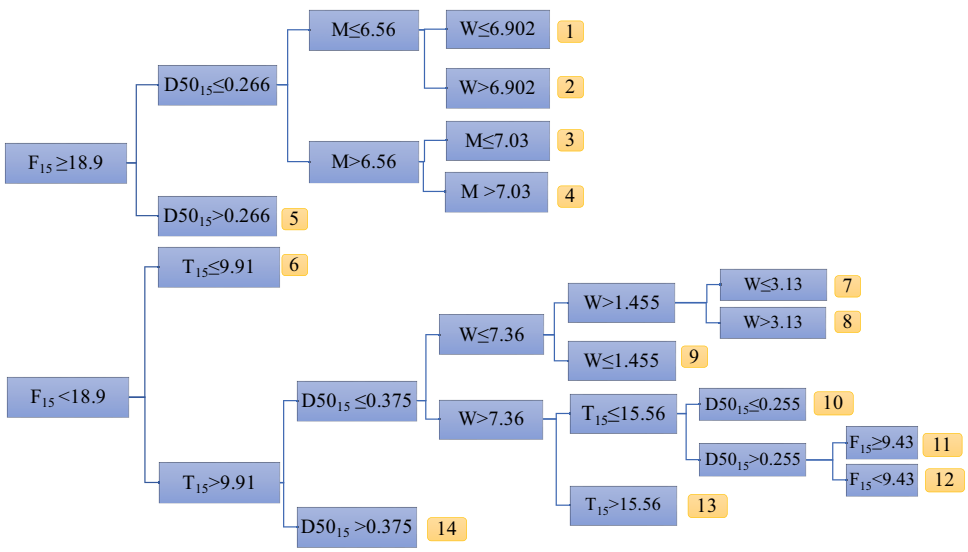

Fig. 9 Conditions of the developed equations

Table. 4 Powers of the presented equations for various conditions

Equation no	Power							
	p1	p2	p3	p4	p5	p6	p7	p8
1	7.3968	−0.6068	0.1828	0.7876	0.443	0.3278	0.0307	−7.4272
2	7.3968	−0.5998	0.3235	0.7876	0.443	0.3158	0.0307	−7.3683
3	9.2932	0.2396	0.1828	0.7876	1.5039	0.2508	0.307	−9.575
4	8.9282	−0.2116	0.3249	0.7876	0.9621	0.3115	0.0307	−8.9754
5	10.4292	−0.5202	0.295	0.7876	−0.13	0.214	0.0307	−10.9624
6	1.5976	−0.1156	0.3474	1.687	−0.352	0.0216	0.264	−4.6855
7	1.9324	−0.1945	0.0742	7.8788	−0.8577	0.1022	0.027	−16.8559
8	1.9324	−0.1945	−1.5074	9.523	−1.2379	0.3506	−0.0713	−18.7795
9	1.9324	−0.1945	0.0742	12.5747	−2.105	0.2658	−0.0713	−26.6901
10	1.9324	−0.1945	0.2252	4.5833	−0.7918	0.1425	−0.1237	−10.414
11	1.9324	−0.1945	0.225	4.8843	−0.7918	0.1425	−0.1237	−11.0209
12	1.9324	−0.1945	0.2252	4.5575	−0.8742	0.1425	−0.1237	−10.3987
13	1.9324	−0.1945	0.333	4.5833	−0.7918	0.1425	−0.1237	−10.516
14	1.9324	−0.295	0.0742	3.1125	−0.9573	0.3607	−0.1183	−7.730

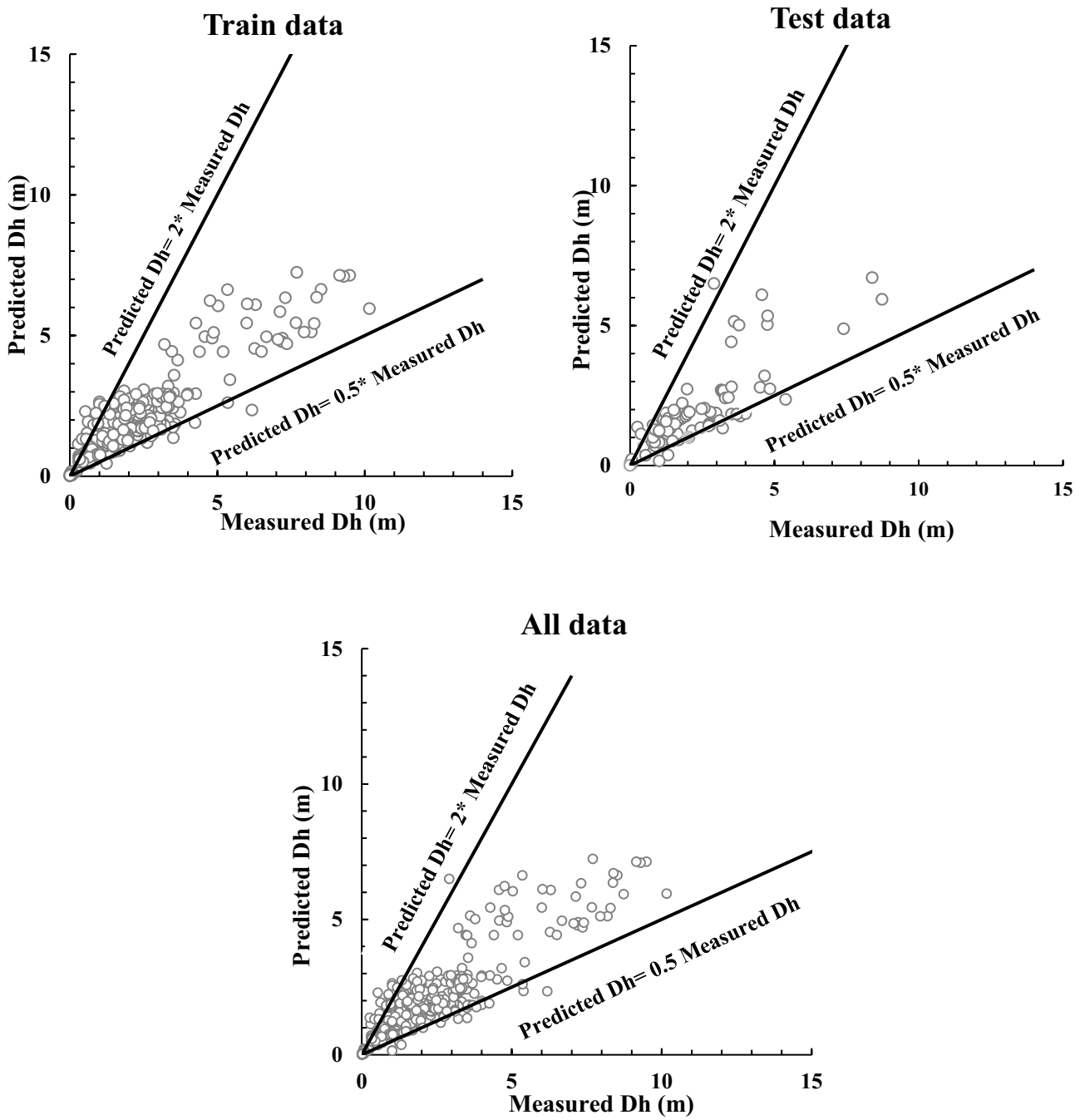

Fig. 10 Predicted versus measured values of lateral ground displacement (Dh) for the (**a**) training, (**b**) testing, and (**c**) all datasets

The standard deviation reduction (SDR) is obtained through the following equation (Pal and Deswal 2009).

$$SDR = sd(S) - \sum \frac{|S_i|}{|S|} sd(S_i) \tag{4}$$

where S is the set of input data at the node, S_i is the subset resulting from the input data division, and sd is the standard deviation. The branching creates a large tree with linear regression models developed on all internal nodes. The subtrees which are not useful for enhancing the accuracy of the model should be removed. Therefore, where the SDR is less

Table. 5 Summary of the results of all models

	Bartlett and Youd (1995)	**Youd et al.** (2002)	Javadi et al. (2006)	**Baziar and Saeedi Azizkandi** (2013)	**Avval and Derakhshani** (2019)	**MT (New model)**
CC	0.76	0.83	0.77	0.86	0.88	0.88
RMSE	1.15	1.08	1.62	0.87	0.88	0.89
MAE	0.8	0.7	1.13	0.63	0.61	0.62

than an anticipated error for a sub-tree, the overgrown tree is trimmed by pruning.

The smoothing step results in the elimination of the interruptions (gaps) created in the pruning step. This involves the integration of the existing model with the new model. The predicted value in each node is filtered in the path from the node to the root and combined with the given value by the linear model as follows:

$$y' = \frac{ny + cx}{n + c} \tag{5}$$

where y' is the estimation moved to the following higher node, y is the estimation moved from the lower node to this node, x is the prediction by the model at this node, n is the number of training data at the lower node, and c is a constant. Hence, the prediction in each node is refined from the node to the root. At last, the MT gives a number of linear multivariable formulae.

Formulation

To create the MT (New model), the whole database including both the sloping ground and free face data was divided into two categories of train and test data comprising 80% and 20% of the data, respectively. These two groups of data are categorized in such a way that they are close to each other in terms of maximum, minimum, average and standard deviation (Table 3). The training dataset was employed in a logarithmic form (log M, log R, log T_{15}, log (100-F_{15}), log ($D50_{15}$+0.1), log W, log S, and log Dh) to develop the new predictive equations using the model tree method. Then, various combinations of governing parameters were utilized to construct the model and the best set of relationships was chosen on the basis of accuracy and simplicity. Finally, the equations were evaluated by the test dataset that was not considered in the process of model creation.

Figure 9 presents the conditions of the best model (branches of the model tree) for choosing the suitable predictive equations. These formulae were created in the form of the sum of the logarithmic parameters as given by Eq. 6:

$$\begin{aligned}\mathrm{Log(Dh)} = {} & \mathrm{p1} \times \mathrm{logM} + \mathrm{p2} \times \mathrm{logR} + \mathrm{p3} \times \mathrm{logT_{15}} + \mathrm{p4} \\ & \times \log\left(100 - \mathrm{F_{15}}\right) + \mathrm{p5} \times \log\left(\mathrm{D50_{15}} + 0.1\right) + \mathrm{p6} \\ & \times \log(\mathrm{W}) + \mathrm{p7} \times \log(\mathrm{S}) + \mathrm{p8}\end{aligned} \tag{6}$$

This equation can be converted to a simple power-law form that is easy to be used in practice. Such a relationship might yield more robust results (Eq. 7) (Juang et al. 2018).

$$\begin{aligned}\mathrm{Dh} = {} & \mathrm{M}^{\mathrm{p1}} \times \mathrm{R}^{\mathrm{p2}} \times \mathrm{T_{15}}^{\mathrm{p3}} \times \left(100 - \mathrm{F_{15}}\right)^{\mathrm{p4}} \\ & \times \left(\mathrm{D50_{15}} + 0.1\right)^{\mathrm{p5}} \times \mathrm{W}^{\mathrm{p6}} \times \mathrm{S}^{\mathrm{p7}} \times 10^{\mathrm{p8}}\end{aligned} \tag{7}$$

The powers of parameters in Eq. 7 for various conditions are summarized in Table 4. It should be noted that in the case of the sloping ground, the free face ratio (i.e., *W*) should be equal to one, and in the case of the free face condition, the ground slope (i.e., *S*) must be equal to unity. It is worth noting that the application of the presented equations in practice, is limited to the range of data employed for developing the model.

Performance of the "MT (New model)"

Accuracy assessment

Figure 10 illustrates the lateral displacement values predicted by the proposed model versus those measured for the

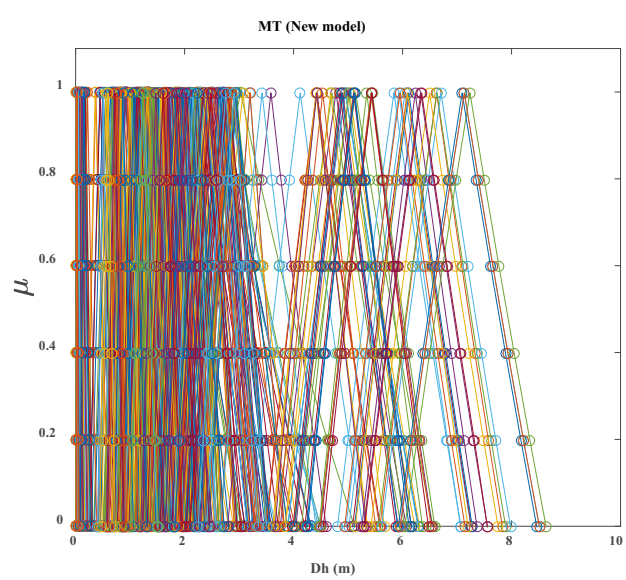

Fig. 11 Membership functions of the developed model responses

train, test, and all data. As shown, the points on the three graphs are approximately between the two lines with the "Predicted Dh = (2 and 0.5) times the Measured Dh", which indicates the good performance of the "MT (New model)" model. This range of 0.5 ~ 2 is a suitable band of accuracy for risk evaluation considering the existing uncertainties (Goh and Zhang 2014; Rezania et al. 2011).

The accuracy of the MT (New model) model was assessed through calculation of the correlation coefficient (CC), root mean square error (RMSE) and mean absolute error (MAE) which are 0.88, 0.89, and 0.62, respectively. The statistical indices of the developed model as well as the models presented by Bartlett and Youd (1995), Baziar and Saeedi Azizkandi (2013), Javadi et al. (2006), Youd et al. (2002), and Avval and Derakhshani (2019) are given in Table 5. As can be seen, the statistical indicators of the new model are almost the same as the MT model of Avval and Derakhshani (2019). Additionally, in contrast to other models, the two models based on model tree and the model of Baziar and Saeedi Azizkandi (2013) have the best overall performance in terms of accuracy.

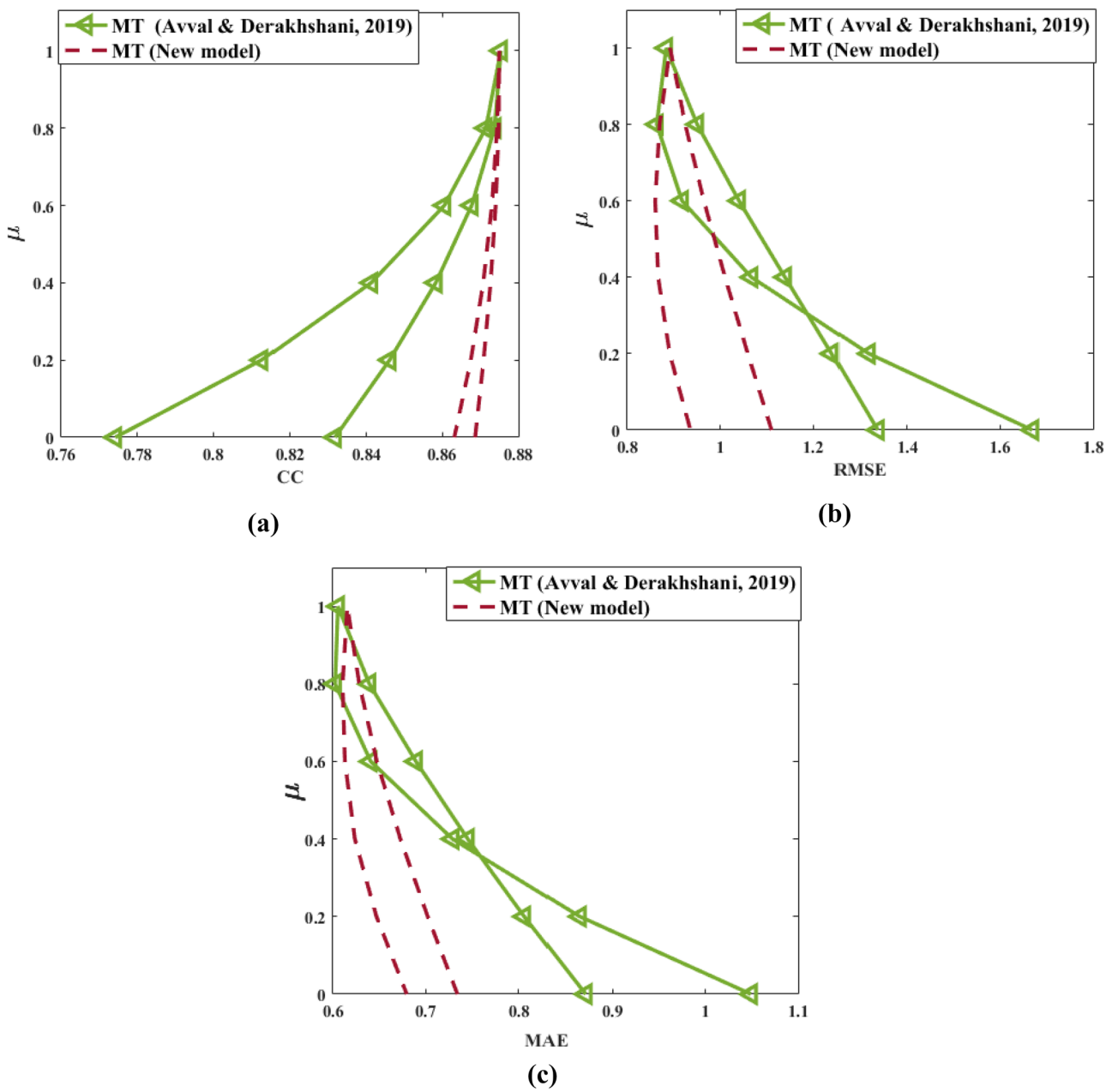

Fig. 12 Membership functions of the MT-based models. (**a**) CC. (**b**) RMSE. (**c**) MAE

Table. 6 Data of the two sample cases of lateral spreading

Condition	Input							Output	
	M	R (km)	T_{15}(m)	F_{15}(%)	$D50_{15}$(mm)	W (%)	S (%)	$Dh_{predicted}$(m)	$Dh_{measured}$(m)
Free face	6.5	2	1.8	30	0.094	8.054	0	0.934	1.03
Sloping ground	7.5	21	10.3	6	0.575	0	0.54	0.961	0.76

Uncertainty assessment

In addition to high accuracy, a good predictive model must also feature high reliability. The fuzzy responses of the MT (New model) to input uncertainties were computed and displayed in Fig. 11. These fuzzy numbers of the lateral spreading can be compared with those of other models depicted in Fig. 4. However, comparing the responses of many data points (also containing various α-cuts) to realize the benefits of the developed model is so complicated. Thus, to make the comparisons easier, the membership functions of the statistical indices (i.e., CC, RMSE, MAE) of the proposed model and the model of Avval and Derakhshani (2019) are plotted in Fig. 12.

As can be seen, the fuzzy indices of CC, RMSE, and MAE of the new model vary in just narrow bounds with maximum widths (corresponding to the fuzzy number supports) equal to 0.005, 0.17, and 0.06, respectively. In other words, the values of statistical indicators at different levels of uncertainty are close to their crisp values. On the other hand, the variations of the fuzzy CC, RMSE, and MAE are up to 0.06, 0.3, and 0.18 for the model of Avval and Derakhshani (2019). Hence, the sensitivity of the CC, RMSE, and MAE membership functions to the uncertainty in the new model is less than those of the other model. Accordingly, it is inferred that the MT (New model) is both accurate and reliable for predicting the amount of lateral spreading.

Implementation examples

To investigate the implementation of the proposed model in engineering practice, we considered two cases of lateral spreading from the database employed for modelling. The input and output data for the selected sample cases of free face and sloping ground conditions are listed in Table 6.

(a) Free face condition: Since $F_{15} \geq 18.9\%$ and $D50_{15} \leq 0.266$ mm and $M \leq 6.56$ and $W > 6.902\%$, condition number 2 is chosen according to Fig. 9 and the values of row 2 of Table 4 are used in Eq. 7 to calculate the displacement. Also, the parameter *S* is equal to 1 in the free face condition. Finally, the lateral spreading is estimated as follows:

$$\begin{aligned} Dh = {} & 6.5^{7.3968} \times 2^{-0.5998} \\ & \times 1.8^{0.3235} \times (100-30)^{0.7876} \\ & \times (0.094+0.1)^{0.443} \times 8.0537^{0.3158} \\ & \times 1^{0.0307} \times 10^{-7.3683} = 0.934(\mathrm{m}) \end{aligned}$$

(b) Sloping ground condition: Condition 14 of Fig. 9 is selected based on the data ($F_{15} < 18.9\%$ and $T_{15} > 9.91$ m and $D50_{15} > 0.375$ mm) to estimate the displacement. The corresponding coefficients from row 14 of Table 4 are substituted in Eq. 7 and because of the sloping ground condition, the value of the parameter *W* equals 1. Accordingly, the lateral spreading is computed as follows:

$$\begin{aligned} Dh = {} & 7.5^{1.9324} \times 21^{-0.295} \\ & \times 10.3^{0.0742} \times (100-6)^{3.1125} \\ & \times (0.575+0.1)^{-0.9573} \times 1^{0.3607} \\ & \times 0.54^{-0.1183} \times 10^{-7.73} = 0.961(\mathrm{m}) \end{aligned}$$

The predicted and measured values of Dh for both cases are compared in Table 6. It can be seen that the "predicted Dh/measured Dh" is in the range of 0.5 ~2.

Conclusions

Predicting the lateral spreading is recognized as being one of the most important topics in the field of earthquake-induced liquefaction. The governing parameters on this phenomenon are subject to uncertainties, and it is necessary to consider the effects of these uncertainties for reliable predictions. In this paper, the influence of uncertainties on the estimations of lateral spreading was evaluated through a fuzzy model created based on fuzzy logic and an evolutionary optimization algorithm. Furthermore, a new predictive model of both high accuracy and good reliability was proposed using the model tree algorithm.

To analyze different available predictive relationships under uncertainty, the inputs were fuzzified via triangular membership functions. The analyses were performed on the well-known database of Youd et al. (2002) used for the development of previous artificial intelligence-based prediction models. The maximum uncertainty of ± 5% was introduced to the membership functions of the input

data with several α-cuts accounting for the uncertainties of ± 4%, ± 3%, ± 2%, and ± 1%. It is noteworthy that the crisp values were also employed for predicting the definite lateral displacement values according to the common traditional approach. The fuzzy responses of the various prediction methods were then obtained by the genetic algorithm (GA) employed to find the extreme values of lateral displacements at each α-cut. The following conclusions can be drawn from this study:

- The vulnerability of different relationships to the influences of uncertainties was evaluated by computing the fuzzy sets of their statistical indicators such as the correlation coefficient (CC), root mean square error (RMSE), and mean absolute error (MAE). The results showed that a slight uncertainty in the inputs leads to substantial uncertainty in the outputs of various relationships. Therefore, a lateral spreading estimate that is utilized for design purposes may not be reliable without considering the uncertainty.
- Through comparison of the fuzzy statistical indices, it was inferred that the recent model of Avval and Derakhshani (2019), i.e., MT, which was superior to its previous models in terms of accuracy, is vulnerable to uncertainty.
- To deal with the drawback of the previously proposed MT model, a new set of formulas was developed to predict the amount of lateral spreading more reliably. The proposed relationships of the "power-law" form were built using the model tree method for both conditions of sloping ground and free face.
- Based on the crisp values of the statistical measures, it was revealed that the new model, called "MT (New model)," has an accuracy similar to that of the MT model. On the other hand, by comparing the fuzzy statistical responses, it was found that the presented set of equations is much less influenced by the uncertain inputs at various α-cuts in contrast to the MT model.
- It is concluded that, when exposed to uncertainty, the new model is capable of making reliable predictions of the lateral spreading which are highly useful in engineering practice.

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